

Security Risks in Wearable IoT Devices

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Abstract—Wearable IoT (Internet of Things) devices are now being implemented in the health care sector and personal lifestyle control, i.e., smartwatches and fitness bands, and medical sensor monitors. These devices are also recording, analyzing, and transferring sensitive data of users like biometric, physiological and location data on a continuous basis. Whichever the benefits they offer wearable objects built around the IoT also have high security and privacy concerns due to the small processing capabilities, the necessity to use wireless connections, and lack of all-encompassing security. The paper will evaluate the security risk of wearable IoT devices, the most common vulnerabilities and attack vectors, and the security measures that have been implemented. A qualitative and analytical approach is adopted in the study that is founded on the recent scholarly articles, technical reports, and case studies. The study demonstrates that wearable IoT devices are especially vulnerable to interception of data, unauthorized access, and poor authentication. Lightweight encryption, secure pairing, and periodic updates of firmware are very useful to reduce potential risks. Lightweight and efficient security setups should be included to create secure wearable IoT ecosystems.

Keywords—Wearable IoT, Cybersecurity, Data Privacy, Lightweight Cryptography, IoT Vulnerabilities

I. INTRODUCTION

Internet of Things (IoT) has transformed the human-technology relationship radically because it enables ordinary objects to accumulate, communicate and react to information with the least human intervention. Wearable technology is now among the fastest-growing spheres of the internet of things devices. Smartwatches, fitness trackers, smart glasses, and medical-grade wearables have become a device used by hundreds of millions of people everywhere to check their health, track their fitness, communicate with others, and even remain productive (and to work) [1][2]. By the year 2024, the number of connected devices using the IoT is more than 18.5 billion around the world, and it is presumed to reach about 14 percent through the years to approximately 21.1 billion by the year 2025 [3].

Such wearable devices are typically packed with a sizeable number of sensors including accelerometers, gyroscopes and heart rate monitors, GPS and temperature sensors that keep recording user data in real time. The information gathered is then transmitted wirelessly, usually using Bluetooth Low Energy (BLE), Wi-Fi to a paired smartphone or directly to cloud servers to be processed and stored [4]. The modern world has embraced its clinical usefulness and ease of usage to the healthcare delivery system, personal wellness management and even in the workplace security as part of these devices.

This mass acceptance has, however, also come along with an alarming assortment of security and privacy consequences. As these devices deal with very sensitive personal

information, such as health-related data, biometric data, coordinates, sleep and physical activities, they are now a profitable target of cyber attackers. Compared to the traditional computing platforms like laptops and desktops, wearable IoT devices have a dire lack of computational power, memory capacity, and battery life. These hardware constraints render it intrinsically challenging to establish traditional security measures, progressive encryption norms, and infiltration event sensors into these devices [1]. The extent of the attack is high: IoT malware attacks increased by 107% in the first half of 2024 [17], and the average cost of a data breach in the world is 4.88 million dollars in 2024 [18].

The complexity of wearable ecosystems in which a device can communicate with smartphones, cloud systems, and other IoT devices, presents several vulnerabilities. Failure in one part of this chain may lead to a possibility of compromising the entire system and providing sensitive information about the users [5]. Due to these increasing concerns, the objective of this paper is to give an in-depth analysis of the security threats posed by wearable IoT devices. It determines the major vulnerabilities and attack vectors, evaluates the current security measures and explains lightweight cryptographic closure, which can be successfully implemented in resource-restrained devices.

TABLE I. GROWTH OF CONNECTED WEARABLE DEVICES WORLDWIDE (2021–2025)

Year	Connected Wearable Devices (Millions)	Year-over-Year Growth
2021	929	+11.3%
2022	1,110	+19.5%
2023	~1,200	+8.1%
2024	~1,300 (est.)	+8.3%
2025	~1,400 (proj.)	+7.7%

^a Sources: IoT Analytics [3]

II. LITERATURE REVIEW

A. Wearable IoT Device Adoption and Growth

The wearable technology market has grown tremendously over the last few years around the world, owing to the rising consumer sensitization on personal health, the development of miniaturized sensor technology, as well as the rising popularity of remote patient monitoring. The highest percentage of this market is taken up by smartwatches and fitness trackers, which are succeeded by the medical-grade wearable devices that are intended to be used in managing chronic diseases as well as in monitoring a patient remotely. Haghi et al. (2017) referred to using wearable devices in

medical IoT and explained that in addition to the ability to monitor the patient in real-time and identify diseases in their initial stages, wearable devices have the disadvantage of gathering the most personal health data ever accessible, and firms must implement advanced data protection systems [2].

B. Security and Privacy Challenges

A number of investigations pointed to the fact that wearable IoT devices have unique security challenges because of their resource-constrained nature. Ching and Singh (2016) noted that wearable devices have a very small CPU, memory and battery space, which constrains the implementation of the conventional security measures like the Advanced encryption standard (AES) using large key size and the public key infrastructure (PKI) [6]. Such limitations compel the manufacturers to use less secure security or in other instances, these manufacturers may decide not to use encryption and authentication.

The systematic review of cybersecurity risk in wearable devices by Ali et al. (2023) revealed that the most prevalent security risks are weak or default authentication credentials, the lack of secure data storage methods, non-encrypted communication channels, and a vulnerability to firmware exploitation [7]. They highlighted that most manufacturers are focusing on quick development processes and cost-cutting instead of focusing on the security issues and ship devices with known vulnerabilities.

Kotz et al. (2016) also paid particular attention to the issue of privacy in wearable health devices, asserting that the ubiquitous and usually non-obtrusive quality of data gathering by these devices creates serious ethical and legal challenges in terms of consent of users, data property, and the possibility of surveillance [8].

C. Attack Models and Threat Vectors

The studies have defined several attack types integrating against a wearable IoT. According to Alrawais et al. (2017), these attacks can be categorized into three layers on the standard IoT architecture perception layer attacks target sensors and hardware, network layer attacks target communication protocols, and application layer attacks target software, APIs, and user interfaces [9]. This stratification is a useful method for considering the different attack surfaces of wearable IoT ecosystems.

Ronen et al. (2017) proved that large-scale IoT attacks are possible, utilizing the vulnerabilities of wireless communication protocols, demonstrated that an attacker can take control over smart devices and distribute malicious firmware updates automatically to the area networks [10]. The 2016 Mirai botnet attack further highlighted the practical nature of the seriousness of IoT security vulnerabilities, whereby hundreds of thousands of infected IoT devices were coordinated to help inflict one of the largest distributed denial-of-service (DDoS) attacks in the history of the internet, taking down major websites and services [11].

D. Lightweight Security Solutions

Considering the resource limitations of wearable IoT devices, scholars have directed more attention towards the development of lightweight security solutions that can maintain security strength and at the same time be computationally efficient. Singh et al. (2017) have written about the use of lightweight block ciphers as PRESENT and SIMON and SPECK to ensure the security of the data on

resource-constrained devices and have shown that the mentioned algorithm can ensure the sufficient level of security with the use of much fewer computational resources than conventional ciphers [12].

The authors Zhang et al. (2019) investigated the application of Elliptic Curve Cryptography (ECC) in IoT systems and proved that ECC provides a very good trade-off between the security level and the computation cost. A 256-bit ECC key has the security of a 3072-bit RSA key, which means that it is faster to compute and it uses significantly less energy, which is why it is an excellent choice in wearable devices [13]. Recently the National Institute of Standards and Technology (NIST) has standardized the Ascon lightweight cipher of authenticated encryption which is planned specifically to be used in constrained environments [14].

III. METHODOLOGY

A. Research Design

The study uses a qualitative and analytical methodology to examine the security threats of wearable IoT devices. The research is structured as a secondary research project, which will conduct a systematic review, analysis, and syntheses of the available academic information on the subject matter. It is structured based on a descriptive-analytical approach, as the security situation of wearable IoT devices is outlined in finer detail, the key attack vectors are classified, and the usefulness of the current countermeasures is critically assessed.

B. Data Collection

The academic databases that were used to gather data in this study are reputable such as the IEEE Xplore, ACM Digital Library, Springer, ScienceDirect and Google Scholar. The search queries were wearable IoT security, IoT vulnerabilities, lightweight cryptography to IoT, wearable device attacks, and data privacy in wearable technology. Inclusion criteria of the chosen studies were as follows: the article should be published within 2013 to 2025, it should dwell upon the analysis of wearable or IoT device security, privacy, vulnerability, and it should be peer-reviewed or published in a reputable journal or magazine and it should be in English. Besides the scholarly literature, there are industry reports and technical advisories of organizations like NIST and SonicWall were also consulted to obtain up-to-date statistical data on the IoT threat landscape.

C. Analysis Approach

The literature gathered was categorized and discussed in 4 main dimensions: the types of wearable devices under investigation (smartwatches, fitness trackers, medical wearables), the types of security threats detected (spoofing, malware, denial-of-service, eavesdropping, man-in-the-middle attacks), the suggested countermeasures and their effectiveness (encryption, authentication protocols, firmware security, intrusion detection), and restrictions and limitations of the existing literature. This multi-dimensional analysis has allowed forming a complete image of the wearable IoT security state of affairs and defining the areas that need a more thorough research.

IV. IOT ARCHITECTURE AND WEARABLE DATA FLOW

Informing about the security threat of wearable IoT devices, it is important to have a detailed understanding of the architecture of the latter, as well as the points at which data

can be stolen or abused. An average wearable IoT system comprises four main layers, which collaborate to gather, relay, process, and store user information.

TABLE II. WEARABLE IOT LAYERED ARCHITECTURE AND DATA FLOW

Layer	Components	Data Flow Direction
Layer 1: Device Layer	Sensors (HR, GPS, Accelerometer, SpO2) → Microcontroller & Local Memory	↓ Wireless Tx (BLE/Wi-Fi)
Layer 2: Communication Layer	BLE / Wi-Fi / NFC Protocols	↓ Short-Range Link
Layer 3: Gateway Layer	Smartphone (Companion App)	↓ Internet (TLS)
Layer 4: Cloud / Server Layer	Cloud Server (Storage & Analytics) → User Dashboard	End Point

Each transition between layers is a potential attack surface, and adversaries can target any link in this chain.

A. Device Layer

The device layer is made up of physical wearable hardware such as accelerometers, gyroscopes, heart rate monitors, SpO2 sensors, GPS modules, microcontrollers, memory units, and communication units such as BLU, Wi-Fi, and NFC. The sensors can measure both physiological and environmental information of the user and this data is momentarily saved in the local memory on the device which is afterwards sent to other systems [2][15]. The hardware units of wearable devices are resource limited because of their small size and the necessity to have a long battery life. This limitation is a direct restriction on the computational power that can be used to run complicated security algorithms, and this inherent constraint is what dictates the overall security problem of wearable IoT devices.

B. Communication Layer

The communication layer will carry out the transfer of data between the wearable device and other parties in the ecosystem. The most used protocol is the Bluetooth Low Energy (BLE) which is used to connect the wearable to a paired smart phone over a short distance. BLE is energy efficient and has been identified to have security vulnerabilities, such as vulnerability to man-in-the-middle (MITM) attacks, eavesdropping and tracking due to persistent Bluetooth addresses [4][16]. Wi-Fi offers better bandwidth and is utilized in certain devices to get direct internet connectivity, although it uses a lot of power, and it opens up more points of attack. Very short-range interaction (less than one centimeter) is done using Near Field Communication (NFC), but some high-end smartwatches have cellular connectivity (LTE/5G) and can be used independently without a paired phone.

C. Gateway and Cloud Layers

A smartphone in most wearable IoT applications serves as an interface between the wearable device and the cloud. The wearable transmits data to the smartphone through Bluetooth, which in turn is locally processed with companion applications, and sent to the cloud servers via the internet. This architecture is another vulnerability point because the security

stance of the smartphone has a direct impact on the overall security of the wearable ecosystem [5]. The remote servers at the cloud layer save and process the data available at wearable devices, undertake analytics, generate health reports, and give users actionable insights using web or mobile dashboards. Although cloud platforms usually utilize strong security protocols (TLS encryption and access control policies), the fact that the data flows between the gateway and the cloud and storing sensitive personal data on third-party infrastructure are long-term issues is still an issue of concern.

In general, the total attack surface (A_{total}) of a wearable IoT ecosystem can be described as the sum of the attack surfaces of each layer:

$$A_{total} = A_{device} \cup A_{communication} \cup A_{gateway} \cup A_{cloud} \quad (1)$$

A_{device} is an example of hardware and firmware vulnerabilities, $A_{communication}$ is vulnerabilities in the wireless protocol (BLE, Wi-Fi), $A_{gateway}$ is vulnerabilities brought by the smartphone and its applications, and A_{cloud} is vulnerabilities to the server infrastructure and stored data. The weakest link in this chain is the determinant of the overall security of the system which explains the significance of defense-in-depth strategies.

V. SECURITY THREATS AND ATTACK VECTORS

The wearable IoT devices are vulnerable to numerous security threats that may threaten the privacy of the user, integrity of the data, and accessibility of the device. This part of the discussion covers the broad groups of attacks that are directed to these devices and the ecosystems around them.

TABLE III. TAXONOMY OF ATTACK VECTORS IN WEARABLE IOT ECOSYSTEMS

Architectural Layer	Attack Vectors
Perception Layer (Device)	Side-Channel Attacks, Firmware Exploitation, Physical Tampering
Network Layer (Communication)	Eavesdropping, Man-in-the-Middle, Denial-of-Service, Device Spoofing
Application Layer (Software & Cloud)	Malware Injection, Unauthorized Access, Data Leakage

A. Eavesdropping and Data Interception

Eavesdropping attack is a type of attack where the information that is being communicated between the wearable device and other elements within the ecosystem is eavesdropped unauthorised. Given that wearable devices are mostly connected via BLE and Wi-Fi, an attacker is likely to intercept the data packets sent through radio communication systems by being within the communication range, with commercial radio equipment [4][16]. When the data is sent either in plaintext or under weak encryption, the attacker will access the contents of the intercepted traffic which can contain sensitive health details, location information and personal identifiers. The use of encryption does not ensure that the communication is secure to packet sniffing and replay attack even when the cryptography is implemented poorly.

It has been shown that even encrypted data can be correctly correlated to the intensity of the user activity as well as the amount of transmitted traffic of BLE to conclude on the type of physical activity the user is currently engaged in i.e.

walking, sitting, or running based on a traffic analysis alone [16]. This is a major privacy threat of such side-channel information leakage in encrypted communication.

B. Device Spoofing and Impersonation

Spoofing attacks involve an attacker that pretends to be an authorized device or object in the IoT topology. Spoofing can take various forms in the context of wearable IoT. The process of device spoofing is when an attacker develops a fake device that impersonates the identity of a legitimate wearable such as copying its MAC address to gain unauthorized access to the smartphone or cloud that it is paired with [9]. Gateway spoofing is the kind of attack in which an attacker configures a rogue access point or a counterfeit smartphone application which the wearable device will connect to automatically and then the attacker is able to intercept all the data relayed and alter it. Spoofing attacks are also highly threatening as they may remain undetected over a long duration, enabling the attacker to keep on harvesting data or adding false information in the medical record system.

C. Malware Injection

Malware attacks of wearable IoT devices can be characterized by the installation of a malicious program on the wearable device, the smartphone that is paired with or the companion app. Wearables are especially susceptible to malware since most of them do not possess the necessary computational power to execute antivirus and application sandboxing, or real-time intrusion detection systems [7]. The major vectors of malware injection are weakened firmware patches, malicious companion programs and exploitation of Bluetooth vulnerabilities.

One of the most informative examples of the IoT malware on the large scale remains the Mirai botnet. Mirai compromised hundreds of thousands of IoT devices by the use of default credentials and familiar firmware vulnerabilities and deployed them to cause a catastrophic DDoS [11].

D. Denial-of-Service (DoS) Attacks

Denial-of-service attacks are used to render a wearable device or its supporting services inaccessible to their authorized user. In the wearable IoT environment DoS can also be implemented by a variety of methods such as wireless channel flooding, resource exhaustion attacks which target the limited processing and memory on the wearable and cloud-level DoS attacks which target the backend servers [9]. In the case of medical wearables, a successful DoS attack may not only have an inconvenience level impact, but may also put actual patient safety at risk, since the loss of real-time health monitoring may postpone life-saving medical interventions.

E. Man-in-the-Middle (MITM) Attacks

In a man in the middle attack, the attacker stands in between the worn-out object and the intended communication partner and clandestinely deciphers, hacks or alters the information that is being transferred. MITM attacks are especially efficient in case of BLE connections, in which no appropriate mutual authentication and cryptographic key exchange protocols are used [4][16]. LE Legacy Pairing mechanism of the BLE protocol that has been adopted by most older wearable devices is reportedly susceptible to passive eavesdropping and active MITM attacks because of its use of a short temporary key (TK) in the pairing mechanism [16].

F. Side Channel Attacks and Firmware Vulnerabilities

Side-channel attacks are a more advanced type of threat which builds on the information leakage caused by the physical properties of the device including power consumption, electromagnetic fields or even timing variations when performing cryptographic operations. Although these attacks need a close proximity and a special equipment, they have been proved to be successful in retrieving the encryption keys in embedded systems [12].

Another common concern is firmware vulnerabilities. Most wearable products are sold with old firmware that already has security vulnerabilities, and their manufactures often fail to update them in time. There are several weaknesses that are common on firmware, such as hard-coded credentials, insecure boot processes, unverified firmware update systems, and buffer overflow vulnerabilities [7].

TABLE IV. SUMMARY OF ATTACK VECTORS, TARGETED LAYERS, AND SEVERITY LEVELS

Attack Type	Targeted Layer	Severity	Primary Risk
Eavesdropping	Communication	High	Privacy breach, data theft
Device Spoofing	Communication	High	Unauthorized access, data manipulation
Malware Injection	Device / Application	Critical	Full device compromise, botnet recruitment
Denial-of-Service	Communication / Cloud	High	Service disruption, patient safety risk
Man-in-the-Middle	Communication	Critical	Data interception and modification
Side-Channel	Device	Medium	Encryption key extraction
Firmware Exploitation	Device	Critical	Persistent unauthorized access

VI. ANALYSIS OF EXISTING SECURITY MECHANISMS

In this segment, the key security measures that have been floated and put into place in response to the risks of wearable IoT devices are assessed.

A. Encryption Techniques

The simplest way of ensuring that wearable IoT systems maintain data confidentiality is through encryption. This domain has received both the symmetric and asymmetric encryption methods. Advanced Encryption Standard (AES) and 128-bit or 256-bit key are commonly believed to be safe and are applicable in most IoT applications. Nonetheless, software-based AES implementations are prohibitive due to their computational overhead to the wearable devices with very limited capabilities. Hardware-accelerated support of AES, increasingly common in current wearable chipsets, does a great deal to address this concern [12].

The throughput-to-energy ratio is a key metric that can be used to assess the computational cost of an encryption algorithm, and it is a key metric to evaluate the wearable IoT deployment:

$$\eta = T / E \quad (2)$$

η is the efficiency of the energy (bits per joule), T (bits/s), and E (J/s). The larger the η value the more energy efficient is the algorithm which is very crucial in the battery powered wearable equipment. Lightweight ciphers like Ascon and PRESENT perform much better on software-only implementation, with the η value being much higher than the AES in constrained microcontrollers [12][14].

The lightweight block ciphers are a good substitute to hardware-accelerated AES where even it is not feasible. Other algorithms like PRESENT, SIMON, SPECK and more recently, an algorithm named Ascon by NIST has been tailored specifically to offer sufficient security and yet with much lower computational, memory, and energy requirements [12][14].

TABLE V. COMPARISON OF ENCRYPTION ALGORITHMS FOR WEARABLE IOT DEVICES

Algorithm	Type	Key Size (bits)	Block Size (bits)	Gate Equivalents (GE)	Suitability for Wearables
AES-128	Block cipher	128	128	~3,400	Feasible with HW acceleration
PRESENT	Block cipher	80 / 128	64	~1,570	Highly suitable
SIMON	Block cipher	64–128	32–128	~1,234	Highly suitable
SPECK	Block cipher	64–128	32–128	~984	Highly suitable
Ascon	AEAD	128	64	~2,570	Highly suitable (NIST standard)

Note: Gate Equivalents (GE) represent the hardware implementation cost; lower GE values indicate more lightweight implementations better suited for constrained wearable devices [12][14].

Elliptic Curve Cryptography (ECC) has been selected as the favored asymmetric encryption method to be implemented in IoT settings as its key sizes are able to offer similar security to RSA but at reduced key sizes that are significantly smaller. An ECC key with a 256-bit provides the same security guarantees as a 3072-bit RSA key, which makes it faster to generate keys, its certificates smaller and less energy is used in the cryptographic processes [13].

TABLE VI. ECC VS. RSA — KEY SIZE AND PERFORMANCE COMPARISON

Security Level (bits)	RSA Key Size (bits)	ECC Key Size (bits)	RSA:ECC Key Ratio	ECC Advantage
80	1,024	160	6.4:1	6× smaller keys
112	2,048	224	9.1:1	9× smaller keys
128	3,072	256	12:1	12× smaller keys
192	7,680	384	20:1	20× smaller keys
256	15,360	512	30:1	30× smaller keys

Source: NIST SP 800-57 Part 1 [14].

B. Authentication Mechanisms

Authentication is used to guarantee that the wearable IoT ecosystem and data are only accessed by authorized devices and users. Access control by use of passwords or PIN is the simplest and most common form of authentication. The implementation of weak or default credentials, or credentials that can be guessed easily, however, often invalidates this strategy and is one of the most frequently used vulnerabilities in consumer IoT devices [7].

MFA is an important enhancement of access control that involves the user presenting two or more independent factors of verification. Wearable devices are also singularly poised to take advantage of the continuous implicit authentication based on the variation of biometric indicators of heart rate, gait patterns recorded by accelerators and gyroscopes and electrodermal activity. Such physiological indicators are naturally hard to fake and their dynamic quality gives them continuous identity verification and not a point-of-entry verification.

In the authentication between devices, i.e., device-to-device and device-to-server, lightweight authentication protocols have been created that utilize computationally efficient cryptographic primitives, including XOR, hash functions, and ECC-based key exchange to create secure session keys with minimal processing costs [13]. These protocols are meant to be used with the limited resource budgets of wearable devices and offer formally verifiable security properties.

C. Secure Communication Protocols

Wearable IoT security is very important in ensuring the integrity and secrecy of data during transmission. The standard protocols of securing internet communication are Transport Layer Security (TLS) and datagram version (DTLS). The complete TLS handshake is however multiproductive and computationally intensive in terms of public key operations and is thus heavy on resource constrained devices. Studies have hence concentrated on the development of improved and pre-shared-key versions of TLS/DTLS that lower the overhead of the handshake, but without compromising the security assurances.

The Bluetooth specification at the device communication level has evolved to accommodate identified weaknesses in security. In BLE 4.2 and subsequent versions, LE Secure Connections were added which use Elliptic Curve Diffie-Hellman (ECDH) key exchange when performing the pairing process to offer significant protection against passive eavesdropping and active MITM attacks [4]. The success of these enhancements is however solely reliant on how well the device manufacturers and companion application developers implement them which is not always a guarantee.

D. Firmware Security and Intrusion Detection

The security requirement of the integrity and authenticity of device firmware is very important. Secure boot systems authenticate the cryptographic signature of the firmware before the machine can start-up, and as such, authenticate that the firmware does not execute a modified or untrustworthy firmware image. Over-the-air (OTA) updates allow manufacturers to remotely apply security patches and bug fixes to devices in the field. Nonetheless, the OTA update functionality should also be resistant to attackers by the

firmware image being digital signed and then verifying through cryptography before installation so to speak before attackers can exploit the update mechanism to deploy malicious code.

The nature of traditional intrusion detection systems (IDS) is usually too resource-intensive to be deployed directly onto wearable gadgets. Instead, researchers have introduced alternative architectures with intrusion detection being done at an edge, usually on the smartphone gateway or a local edge server, but not on the wearable device per se [19]. Anomaly detection detectors using machine learning have likewise proven to be effective, with all models being trained according to the normal device behavioral patterns, and network traffic profiles, to identify anomalies that can signal a security event [20].

VII. RESULTS AND DISCUSSION

A. Key Findings

The detailed review and analysis presented as part of this work provides several important conclusions concerning the security situation of wearable IoT devices. The general risk attitude of wearable IoT devices can be measured in the form of the commonly applied risk assessment formula commonly used in the practice of information security:

$$R = P(T) \times V \times I \quad (3)$$

R is the overall score of the system risk, P(T) is the likelihood of occurrence of a particular threat, V is the vulnerability state of the system (between 0 and 1), and I is the level of impact in case of successful exploiting. In the case of wearable IoT devices, V is generally high, because the resource limitations that restrict the security implementation scales, P(T) is growing yearly as it is shown that the number of IoT attacks has increased by 107% in 2024 [17], and I is severe, as health and biometric data are sensitive information.

To begin with, data interception is one of the most widespread and the most available threats to the security of wearable IoT. The prevalence of BLE to communicate with devices and the reality that a large fraction of devices use older versions of protocols or do not use the only offered security measures properly expose a large percentage of wearable transmissions of data to eavesdropping and traffic analysis [4][16].

Second, in the wearable IoT, weak authentication protocols are still enabling unauthorized access. The default credentials that many devices persistently store and the lack of multi-factor authentication in the majority of devices cause a system-wide vulnerability that attackers actively take advantage of [7].

Third, wearable technologies have an inherent tension between security and usability due to their resource limits. The conventional encryption and authentication protocols are not usually computationally efficient on these platforms and require the implementation of lightweight security options [1][12].

Fourth, light cryptography has been shown to be really promising. Present, SIMON, SPECK, and the NIST standardized Ascon are algorithms that offer sufficient security levels appropriate to the sensitivity of wearable information and can be run within the limited resource

constraints of these devices. The protocols based on ECC also increase this power by providing effective asymmetric cryptographic functions [12][13][14].

Fifth, a firmware security is a rather unaddressed field of the wearable IoT industry. Most manufacturers do not avail security patches in a timely manner, and the popular firmware vulnerabilities like hard-coded credentials and insecure boot processes continue to exist [7].

Sixth, there is no one security measure that can secure wearable IoT devices in the whole range of threats identified. The comprehensive protection of a multi-layered defense approach, aligning with encryption, strong authentication, secure communications protocols, firmware integrity verification and anomaly environment analysis, is necessary [7].

B. Discussion

The results of this paper create a physical image of a wearable IoT sector that grows at an accelerated rate, but its security maturity has failed to match its market growth. According to the study by Ali et al. (2023), most manufacturers think about quick developmental processes and lower costs instead of security and ship the devices that they know are vulnerable [7]. Lack of a standard widely used security standard specifically on wearable IoT devices implies that there is a high degree of variation in security implementations across manufactures and product lines.

The economic cost of not implementing wearable IoT security is significant and can be presented in the terms of the Annualized Loss Expectancy (ALE) model that is widely applicable in information security economics:

$$ALE = ARO \times SLE \quad (4)$$

and ALE is the Annualized Loss Expectancy ARO is the Annualized rate of Occurrence and SLE is the Single Loss Expectancy. The ALE of a company that uses the wearable IoT is rising significantly with the data breach average of \$4.88 million per organization in 2024 [18], and the annual rise in IoT attack numbers, up by 107 percent [17].

There are new technologies that can be used as a way of improvement. IoT security has also suggested AI-based anomaly detection systems, which have proven to be effective in detecting unusual behavior of a device [20]. Nevertheless, the implementation of such technologies in real scenarios and on resource constrained wearable gadgets is an actively researched topic.

VIII. CHALLENGES AND FUTURE DIRECTIONS

A. Current Challenges

Nevertheless, despite the significant advancements in the research of IoT security, there are still several challenges to the effective protection of wearable IoT devices, which remain. The variety of wearable devices, which includes devices with extremely diverse hardware capabilities, operating systems, and communication protocols, is incredibly challenging to come up with universal security solutions [1]. Scalability is another issue: as the number of connected wearable devices keeps its dramatic increase toward the estimated 21.1 billion connected IoT devices by 2025 [3], security solutions must be scalable with no performance penalty that are unacceptable. Energy efficiency

is another key design challenge, particularly for medical wearables that must be used for days or weeks without being recharged. The user education is as well a major obstacle with a large portion of the consumers being unaware of the security risks of their wearable technologies and not taking simple precautions, such as implementing updates to their firmware, using a strong password, or avoiding non-secure public Wi-Fi connections [7]. Last but certainly not least, the regulatory environment of IoT security is still dynamic and there are mostly no enforceable, explicit guidelines which aim at defining the security considerations of wearable-based devices in most jurisdictions.

B. Future Directions

Resting on the results of the current study, it is possible to find some directions of the future research and industry activity. To start with, a sense of urgency surrounds the need to develop coherent security frameworks that are specifically tailored to wearable IoT devices that may approach security holistically, at both architectural levels. Second, it is necessary to promote the introduction of the artificial intelligence and machine learning into wearable security architecture to allow detecting threats and reacting to them in real-time. Third, as quantum computing technology advances, the development of lightweight post-quantum cryptography algorithms for IoT deployment is becoming increasingly important. Fourth, privacy-preserving methods of data processing, including federated learning and homomorphic encryption, should be considered as the mechanisms of meaningful health analytics that allow keeping sensitive user data decentralized and encrypted. Fifth, there has to be industry-wide standardization, supported by organizations like NIST, IEEE, and ISO, that provides minimum security requirements and best practices that wearable IoT manufacturers need to follow.

IX. CONCLUSION

The wearable IoT devices have entered the world of modernity, and they provide feasible contributions to the healthcare monitoring, fitness management, personal productivity, and chronic disease management. But at the same time, the fast adoption of such devices has further presented a very widespread and growing attack surface that cybercriminals are actively attacking as witnessed by the 107% increase in IoT malware attacks in 2024 [17]. The resource-constrained characteristics of wearable hardware become a key determinant of the security challenges and force the application of the traditional security tools and the This paper has discussed in detail the risks of security of wearable IoT devices such as eavesdropping, device spoofing, malware injection, denial-of-service attacks, man-in-the-middle attacks, side-channel attacks, and vulnerabilities of firmware. The discussion of the currently used security mechanisms shows that the use of such solutions as lightweight cryptographic algorithms (PRESENT, SIMON, SPECK, Ascon), Elliptic Curve Cryptography, multi-factor and biometric authentication, and secure communication protocols (BLE Secure Connections, CoAP with DTLS) might be combined as effective security measures when they are well-implemented and maintained. The manufacturers are however not consistent in adopting these measures and firmware management, user education and enforcement of the measures all have major gaps.

The provision of security in wearable IoT devices is not only a technical necessity but a condition of ensuring privacy

of users, integrity of health data and continued trust of the population on wearable technology. Effective security architectures are crucial in the construction of secure and reliable wearable IoT ecosystems through integrating lightweight security architectures. In the future, a joint venture involving equipment manufacturers, security professionals, jurisdiction, and users will be required to bridge the difference between the rate of wearable IoT innovation and the maturity of its security infrastructure.

X. EMERGING SECURITY TECHNOLOGIES FOR WEARABLE IoT

New technologies have been investigated by the recent research to enhance the security of wearable IoT ecosystems greatly. The use of blockchain-based frameworks to achieve decentralized device authentication and the exchange of secure data is one of the promising methods. The blockchain will be able to offer unalterable logs of the communication of the device and make sure that only authorized nodes will be involved in the wearable network. It has been proposed by researchers that lightweight blockchain frameworks along with edge computing can be used to address the computational constraints that are generally attached to blockchain applications in IoT settings.

The other upcoming practice is the implementation of artificial intelligence (AI) and machine learning to detect anomalies in wearable devices networks. Machine learning models may be used to identify abnormal activities by studying the behavior patterns of devices like frequency of communication, sensor usage patterns, and attempted authentication. Such systems are capable of detecting possible attacks like spoofing, botnet actions or unsanctioned access attempts on real-time. With the adoption of AI models, they are becoming more and more a part of edge gateways and mobile devices that can connect with wearables.

Secure hardware modules have also been noted to be gaining traction as a wearable security enhancement tool. Hardware-level encryption key and sensitive data protection can be done by Trusted Platform Modules (TPM), Physically Unclonable Functions (PUF), and secure enclaves. These technologies also minimize the chances of key extraction attacks as well as device tampering by keeping cryptographic keys in secure hardware environments.

XI. RECOMMENDATIONS FOR SECURE WEARABLE IoT DESIGN

To curb security vulnerability in wearable IoT ecosystems, manufacturers and system designers need to consider several best practices. To begin with, security should not be a post-design consideration when it comes to the development of a device. Secure-by-design concepts urge developers to consider threat models, attack surfaces, and risk mitigation measures throughout the early stages of product design. Second, wearable devices, smartphones, and cloud platforms should have strong authentication between them. Multi-factor authentication, certificate-based authentication, and secure pairing protocol can be of great help in minimizing the risk of unauthorized access. Constant updates of the firmware and over the air (OTA) patching of new vulnerabilities should also be introduced.

Third, privacy of data should be taken seriously. Wearable are devices that gather very sensitive health and biometric information, and this must always be encrypted during the

process of transit and rest. Differential privacy and secure data aggregation are privacy preserving mechanisms that can further decrease the possibility of user identification in aggregated data.

Last but not least, regulation compliance is very important in wearable security. Such frameworks as the General Data Protection Regulation (GDPR) and medical regulations like HIPAA offer rules concerning the safety of sensitive health information. Companies that adhere to these frameworks have high chances of adopting effective data protection.

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